

Comprehensive Analysis of Trader Performance Under Bitcoin Market Sentiment

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Abstract

Cryptocurrency markets are uniquely driven by investor psychology, where emotions such as fear and greed often dominate rational decision-making. This project presents a comprehensive quantitative study of the relationship between market sentiment and trader performance by integrating the Bitcoin Fear & Greed Index with historical trader execution data from Hyperliquid.

Through a structured analytical framework consisting of Sentiment Regime Analysis, Trader Behavior Clustering, and Sentiment Transition Event Studies, this work uncovers behavioral patterns, reaction lags, and risk inefficiencies embedded in trader decision-making. The findings demonstrate that trader performance is highly sentiment-dependent, heterogeneous across trader archetypes, and particularly vulnerable during sentiment regime transitions. The results provide actionable insights for sentiment-aware trading strategies, behavioral risk management, and future predictive modeling systems.

1. Introduction

Financial markets are not purely mechanical systems; they are deeply influenced by human psychology. In cryptocurrency markets, this influence is amplified due to:

- High volatility
- 24/7 trading
- Retail-dominated participation
- Rapid information diffusion

The Bitcoin Fear & Greed Index has emerged as a widely accepted proxy for measuring collective market emotion. However, while sentiment indicators are commonly observed, they are rarely integrated with actual trader-level execution data to study real behavioural outcomes.

Problem Statement

Most trading analyses focus on price action or indicators in isolation, ignoring:

- How different traders behave under the same sentiment
- Whether traders anticipate or react to sentiment changes
- How behavioral biases translate into profit or loss

Objective

The objective of this project is to:

1. Quantify how trader performance varies across sentiment regimes
2. Identify distinct trader behaviour archetypes
3. Analyse trader reactions to sentiment transitions

4. Convert psychological signals into measurable, actionable insights

2. Dataset Description

2.1 Bitcoin Fear & Greed Index Dataset

This dataset captures daily market sentiment using a composite score derived from volatility, momentum, social media activity, and market dominance.

Columns:

- timestamp: UNIX timestamp
- value: Numerical sentiment score (0–100)
- classification: Categorical sentiment label
- date: Calendar date

Sentiment Classes:

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

This dataset represents the **emotional state of the overall market**.

2.2 Hyperliquid Historical Trading Dataset

This dataset contains granular, trade-level execution data from real trader accounts.

Key Columns:

- Account: Trader wallet address

- Coin: Traded asset
- Execution Price: Trade execution price
- Size Tokens, Size USD: Trade size
- Side: Buy/Sell
- Timestamp IST: Execution time
- Closed PnL: Realized profit or loss
- Fee: Transaction cost

This dataset reflects actual trader behavior, not theoretical strategies.

3. Data Preprocessing & Integration

To ensure analytical accuracy, extensive preprocessing was performed:

1. Date Standardization

- Converted sentiment dates and trade timestamps into comparable datetime formats.

2. Trade Filtering

- Retained only trades with valid Closed PnL.

3. Daily Alignment

- Aggregated trades by trading date to match sentiment granularity.

4. Data Integration

- Merged both datasets on `trade_date`.

The resulting dataset (`merged_df`) contains:

- Trader identity
- Trade execution details
- Profitability metrics

- Corresponding daily market sentiment

4. Technique 1: Sentiment Regime Performance Analysis

4.1 Objective

To evaluate how trader performance varies under different market sentiment regimes.

4.2 Methodology

Trades were segmented based on sentiment classification. For each regime, the following metrics were computed:

- Total number of trades
- Average and median PnL
- Win rate
- Average trade size (USD)
- Average transaction fees

4.3 Visual Analysis

The following figures were generated:

- Average Trader PnL Across Market Sentiment Regimes
- Trader Win Rate by Market Sentiment
- Average Trade Size (USD) by Sentiment Regime

4.4 Key Findings

- Trader profitability is strongly sentiment-dependent.
- Greed and Extreme Greed regimes show higher risk exposure but weaker consistency.

- Fear regimes encourage more disciplined and selective trading.
- Neutral sentiment acts as a baseline performance state.

4.5 Interpretation

Sentiment regimes influence risk appetite, not just market direction. High optimism leads to aggressive positioning, while fear enforces caution often improving trade quality.

5. Technique 2: Trader Behavior Profiling & Clustering

5.1 Objective

To identify hidden trader archetypes based on behavior rather than raw performance.

5.2 Feature Engineering

Trades were aggregated at the trader level to compute:

- Trade frequency
- Average PnL
- PnL volatility
- Win rate
- Average trade size
- Average sentiment exposure

Features were standardized to ensure fair clustering.

5.3 Clustering Approach

- K-Means clustering was applied.

- The Elbow Method was used to select the optimal number of clusters.
- Traders were grouped into distinct behavioral profiles.

5.4 Visual Analysis

Generated figures include:

- Elbow Method for Optimal K
- Average PnL by Trader Cluster
- Cluster-wise PnL Across Market Sentiments

5.5 Identified Trader Archetypes

Clusters typically represent:

- High-risk, overconfident traders
- Disciplined, low-risk traders
- High-frequency scalpers
- Conservative, low-activity traders

5.6 Key Insights

- Trader behavior matters more than trade volume.
- Disciplined clusters outperform consistently.
- Risk-heavy clusters suffer disproportionately during Greed regimes.

6. Technique 3: Sentiment Transition & Event-Based Analysis

6.1 Objective

To analyze trader behavior before and after sentiment regime changes.

6.2 Methodology

- Sentiment was encoded numerically.
- Regime changes were treated as events.
- A ± 3 -day event window was analyzed.
- Transitions were classified as:
 - Positive (Fear $\rightarrow$ Greed)
 - Negative (Greed $\rightarrow$ Fear)

6.3 Visual Analysis

- **Trader PnL Before vs After Sentiment Transitions**
- **Trade Size Behavior Around Sentiment Transitions**

6.4 Key Findings

- Traders react after sentiment changes.
- Fear $\rightarrow$ Greed transitions lead to premature risk-taking.
- Greed $\rightarrow$ Fear transitions cause delayed exits and sharp losses.
- Transition windows are the highest-risk periods.

7. Cross-Technique Insights

By combining all three techniques:

- Sentiment determines *when* risk is taken.
- Trader behavior determines *who* succeeds.

- Transitions determine *where* losses concentrate.

8. Practical Applications

- Sentiment-aware trading strategies
- Dynamic risk and position sizing
- Trader profiling and capital allocation
- Feature engineering for ML-based prediction systems
- Automated sentiment-triggered safeguards

9. Limitations

- Sentiment index is market-wide, not asset-specific.
- No order-book depth data included.
- Leverage effects not explicitly modeled (future scope).

10. Conclusion

This project demonstrates how market sentiment directly influences trader behavior and performance in cryptocurrency markets. By integrating the Bitcoin Fear & Greed Index with real trading data from Hyperliquid, the analysis shows that trader outcomes are highly dependent on emotional market regimes rather than price movements alone. Periods of Greed are associated with increased risk-taking and larger trade sizes, but not necessarily better profitability, while Fear-driven markets tend to promote more disciplined and consistent trading behavior.

The study further highlights that traders are not homogeneous; distinct behavioral archetypes exist, and disciplined traders consistently outperform high-risk participants across sentiment conditions. Additionally, sentiment transition periods emerge as the most critical

phases, where traders often react late, leading to amplified losses during market reversals.

Overall, this work establishes sentiment as an actionable and measurable signal. By combining sentiment analysis with trader-level behavioral insights, the framework provides a strong foundation for sentiment-aware trading strategies, improved risk management, and future predictive modeling in cryptocurrency markets.